

INTOXICATE Analysis: Sex Differences and Dysrhythmia Validation

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##  
## === TRAIN/TEST SPLIT SUMMARY ===  
  
## Training set (OLD - published study): n = 101  
  
## Test set (NEW - validation): n = 73  
  
## Total: n = 174  
  
## =====
```

PART 1: DEMOGRAPHICS

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##
## === DEMOGRAPHICS FOR ABSTRACT (OVERALL) ===

## Total: n = 174

## Females: 84 (48%), median age 28 [IQR 20-44]

## Males: 90 (52%), median age 38 [IQR 28-67]

## =====

##
## === DEMOGRAPHICS BY DATASET (FOR ABSTRACT) ===

##
## ORIGINAL DATASET (n=101):

##   Females: 54 (53%), median age 28 [IQR 17-43]

##   Males: 47 (47%), median age 31 [IQR 23-44]

##
## NEW DATASET (n=73):

##   Females: 30 (41%), median age 29 [IQR 27-64]

##   Males: 43 (59%), median age 63 [IQR 30-76]

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PART 2: SEX DIFFERENCES IN EXPOSURE PATTERNS

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##
## === SEX-EXPOSURE ANALYSIS ===

## Chi-square p-value: 0.014

## Antidepressants: 16 female (76%), 5 male (24%)

## OR (manual calc): 4.00

## OR (Fisher): 0.25 (95% CI: 0.07 - 0.77)

## Fisher p-value: 0.009

## Bonferroni adjusted p: 0.037

##
## CO/As/CN (occupational): 6 female (32%), 13 male (68%)

## OR for males: 2.19

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Table 1. Distribution of Exposure Categories by Sex

Exposure	Female	Male	Total	P-value
Alcohol	3 (30)	7 (70)	10	0.014
Analgesic	15 (62)	9 (38)	24	
Antidepressants	16 (76)	5 (24)	21	
CO, As, CN	6 (32)	13 (68)	19	
Combination	19 (40)	29 (60)	48	
Sedatives	7 (70)	3 (30)	10	
Street Drugs	6 (32)	13 (68)	19	
Unknown	12 (52)	11 (48)	23	
Total	84 (48)	90 (52)	174	

PART 2B: ADOLESCENT SUBGROUP ANALYSIS (Ages 12-19)

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##
## === ADOLESCENT SUBGROUP (Ages 12-19) ===

## Total adolescents: n = 34

## Females: 21 (62%), median age 15 [IQR 13-16]

## Males: 13 (38%), median age 16 [IQR 15-16]

##
## Chi-square p-value: 0.127

## Antidepressants: 5 female (83%), 1 male (17%)

## OR (manual calc): 3.75

## OR (Fisher): 0.28 (95% CI: 0.01 - 2.96)

## Fisher p-value: 0.370

## Bonferroni adjusted p: 1.000

##
## Analgesics: 7 female (78%), 2 male (22%)

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Table 2. Adolescent Subgroup: Distribution of Exposure Categories by Sex (Ages 12-19)

Exposure	Female	Male	Total	P-value
Alcohol	0 (0)	3 (100)	3	0.127
Analgesic	7 (78)	2 (22)	9	
Antidepressants	5 (83)	1 (17)	6	
CO, As, CN	2 (67)	1 (33)	3	
Combination	1 (100)	0 (0)	1	
Street Drugs	3 (38)	5 (62)	8	4
Unknown	3 (75)	1 (25)	4	
Total	21 (62)	13 (38)	34	

PART 3: PREDICTION MODEL ANALYSIS (Train/Test Validation)

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##
## === OUTCOME MODELING SAMPLE SIZES ===

## Training: n=101 (ICU cases: 19, 18.8%)

## Test: n=66 (ICU cases: 9, 13.6%)

## =====

##
## === SEX COEFFICIENT (Training Data) ===

## Beta: 1.795

## SE: 0.739

## P-value: 0.015

## LRT p-value: 0.009

## =====

##
## === TEST SET PERFORMANCE (Threshold = 0.5) ===

## Model WITHOUT Sex:

##   Sensitivity: 44.4%

##   Specificity: 86.0%

##   AUC: 0.663

##
## Model WITH Sex:

##   Sensitivity: 44.4%

##   Specificity: 87.7%

##   AUC: 0.602

##
## DeLong test p-value: 0.308

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##
## === TEST SET PERFORMANCE (Training-Optimal Thresholds) ===

## Threshold WITHOUT sex (from training): 0.209

## Threshold WITH sex (from training): 0.132

##
## Model WITHOUT Sex:

##   Sensitivity: 55.6%

##   Specificity: 73.7%

##
## Model WITH Sex:

##   Sensitivity: 66.7%

##   Specificity: 47.4%

## =====

##
## === TEST SET PERFORMANCE (Same Threshold for Both) ===

## Shared threshold (from model without sex): 0.209

##
## Model WITHOUT Sex:

##   Sensitivity: 55.6%

##   Specificity: 73.7%

##
## Model WITH Sex:

##   Sensitivity: 55.6%

##   Specificity: 64.9%

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Summary Table

Metric	Without Sex	With Sex
Threshold = 0.5		
Sensitivity	44.4%	44.4%
Specificity	86.0%	87.7%
AUC	0.663	0.602
Training-Optimal Threshold		
Sensitivity	55.6%	66.7%
Specificity	73.7%	47.4%
Shared Threshold		
Sensitivity	55.6%	55.6%
Specificity	73.7%	64.9%

CONCLUSION

Key findings:

1. **Sex-Exposure Association (Full Cohort):** Chi-square $p = 0.014$. Antidepressants: 76% female (OR 4.0, Fisher $p = 0.009$, Bonferroni adjusted $p = 0.037$).
2. **Adolescent Subgroup (Ages 12-19):** $n = 34$. Chi-square $p = 0.127$. Antidepressants: 83% female (OR 3.8, Fisher $p = 0.370$, Bonferroni adjusted $p = 1.000$).
3. **Sex in Prediction Model:** LRT $p = 0.009$. Sex coefficient = 1.80 ($p = 0.015$).
4. **Test Set Performance:** AUC without sex = 0.663, AUC with sex = 0.602 (DeLong $p = 0.308$).

Recommendation for abstract: Report sensitivity/specificity as primary metrics. Including biological sex in the model changed sensitivity from 44% to 44% and specificity from 86% to 88%. Adolescent subgroup findings can be added to Results section.